

Complex Analysis PhD Exam, September 2011

Do all 9 problems. The symbol U always denotes the unit disc: $|z| < 1$; $\mathbb{C}$, the complex plane; Ω , an open region in $\mathbb{C}$; and $\Re z$, the real part of z .

1. Evaluate the integral $\int_0^\infty \frac{(\log x)^3}{1+x^2} dx$.
2. Let f_n , $n = 1, 2, 3, \dots$, be a sequence of one-to-one analytic functions on Ω . If f_n converges uniformly to f on any compact subset of Ω , prove that f is either constant or one-to-one. Show by examples that both conclusions can occur.
3. Let F be a family of analytic functions f on U such that $\Re f > 0$ and $f(0) = 1$. Prove that F is a normal family. Can the condition $f(0) = 1$ be omitted? Justify your answer.
4. Suppose f is a conformal mapping of U onto a square with center at 0 and $f(0) = 0$. Prove that $f(iz) = if(z)$. If $f(z) = \sum_0^\infty c_n z^n$, prove that $c_n = 0$ unless $n - 1$ is a multiple of 4.
5. Let R be a rational function such that $|R(z)| = 1$ if $|z| = 1$. Prove that R is of the form $R(z) = cz^m \prod_{n=1}^k \frac{z-a_n}{1-\bar{a}_n z}$, where c is a constant with $|c| = 1$, m is an integer and $a_1, a_2, \dots, a_k$ are complex numbers such that $a_n \neq 0$ and $|a_n| \neq 1$.
6. Let f be analytic in Ω . If f has no zeros in Ω , prove that $\log |f|$ is harmonic in Ω .
7. Suppose that $f : \mathbb{C} \rightarrow \mathbb{C}$ is a continuous function and f is analytic everywhere except possibly on $[-1, 1]$. Prove that f is an entire function.
8. Suppose f and g are both analytic in U , and neither of them has a zero in U . If $\frac{f'}{f}(1/n) = \frac{g'}{g}(1/n)$ for $n = 2, 3, 4, \dots$, find a simple relation between f and g .
9. Construct a conformal map $w = f(z)$ which maps U onto the angular region $A : |\arg(w)| < \alpha$ and $f(0) = 1$, where $0 < \alpha < \pi$.